

Vehicle Check Backend

Complete System Documentation

UK Vehicle History Report API

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1. System Overview

This NestJS backend provides a comprehensive UK vehicle history check service. It integrates 5 different data sources with an intelligent 4-step fallback strategy to deliver full vehicle reports matching the quality of premium services like Check Car Details.

Technology Stack

- NestJS (TypeScript) with Mongoose ORM
- MongoDB Atlas for database
- Stripe for subscription payments with webhook
- JWT-based authentication (access + refresh tokens)
- pdfkit for PDF report generation
- 5 external APIs: DVLA, CarTax RapidAPI, UKVD, CheckCarDetails, DVSA MOT

User Tiers

The system supports two user tiers with different access levels:

Feature	Free User	Paid/Subscribed User
Vehicle Check	Full JSON data (all 22 sections)	Full JSON data (all 22 sections)
PDF Report	Not available (403)	Downloadable PDF report
Report Storage	Saved to MongoDB	Saved to MongoDB
DVLA Key Used	Free DVLA key	Paid DVLA key (when fixed)

2. API Architecture — 4-Step Data Flow

When a user submits a registration number (VRM), the backend executes a 4-step data aggregation pipeline. Each step enriches the report with additional data. The final result is a deeply-merged JSON object with all available fields.

Step-by-Step Flow

STEP 1 — CheckCarDetails API (PRIMARY)

Calls 7 data points in parallel: `vehicleregistration`, `ukvehicledata`, `Vehiclespecs`, `carhistorycheck`, `mot`, `mileage`, `vehiclevaluation`. Provides the most comprehensive data when live key is active.

STEP 2 — CarTax RapidAPI (FALLBACK)

Only called if Step 1 did not return a vehicle make. Uses RapidAPI endpoint (`uk-vehicle-data1`) for comprehensive vehicle data including stolen/write-off/finance checks, mileage, valuation, and keepers history.

STEP 3 — UKVD API (ENRICHMENT)

Always called. Fills in technical specifications that other APIs may miss: performance data (BHP, torque, 0-60), fuel economy (urban/extra-urban/combined MPG), dimensions (height, width, length, weight), engine data (cylinders, aspiration, fuel system), and SMMT details.

STEP 4 — DVLA Vehicle Enquiry API (ENRICHMENT)

Always called. Fills in official DVLA data: tax status, MOT status, make, colour, fuel type, engine capacity, CO2 emissions, year of manufacture, V5C issue date, and export marker. Uses free DVLA key (paid key currently has auth issues).

Deep Merge Strategy

Each step's parsed result is merged into the master report using a deep merge function. Fields already populated by earlier steps are never overwritten — only empty/null fields get filled. This means the highest-priority source (Step 1) always takes precedence.

3. All 5 Data Sources — Detailed

3.1 CheckCarDetails API (CCD)

Base URL: `https://api.checkcardetails.co.uk/vehicledata/{DataPoint}`

Auth: Query param `apikey={KEY}`

Env Variable: `CCD_TEST_API_KEY` (test) / `CCD_LIVE_API_KEY` (production)

File: `src/app/helpers/checkCarDetailsAPI.ts`

Data Points called in parallel:

Data Point	Returns	Test Key Status
vehicleregistration	DVLA registration, tax, MOT status	WORKING
mot	MOT history, test results	WORKING
mileage	Mileage history, analysis	WORKING
ukvehicledata	Full vehicle data, specs, SMMT	NEEDS LIVE KEY
Vehiclespecs	Performance, dimensions, fuel economy	NEEDS LIVE KEY
carhistorycheck	Stolen, finance, write-off, salvage	NEEDS LIVE KEY
vehiclevaluation	Market valuation data	NEEDS LIVE KEY

Note: Test key only works with VRMs containing the letter 'A'. Premium data points (ukvehicledata, Vehiclespecs, carhistorycheck, vehiclevaluation) return 403 with test key — require CCD Live API key at £20/month.

3.2 CarTax RapidAPI (uk-vehicle-data1)

Base URL: `https://uk-vehicle-data1.p.rapidapi.com/cartax.api.v1.Public/GetInitialReport`

Auth: `x-rapidapi-key` header

Env Variable: `CARTAX_API_KEY`

File: `src/app/helpers/carTaxAPI.ts` — `callCarTax()` / `parseRapidApiResponse()`

Provides comprehensive vehicle data including mileage analysis, stolen/write-off/finance checks, keeper history, plate changes, and basic specifications. This is the primary fallback when CCD is unavailable.

3.3 UKVD (UK Vehicle Data)

Base URL: `https://uk1.ukvehicledata.co.uk/api/datapackage/{Package}`

Auth: Query param `auth_apikey={KEY}`

Env Variable: `UKVD_API_KEY` (currently: `0DBC4CC1-AA7D-4315-9204-2AE33545AE6B`)

File: `src/app/helpers/carTaxAPI.ts` — `callUkvd()` / `parseUkvdData()`

Provides the richest technical specification data: nested performance objects (Power.Bhp, Torque.Nm, MaxSpeed.Mph, Acceleration.ZeroTo60), fuel consumption (Combined.Mpg, UrbanCold.Mpg, ExtraUrban.Mpg), full dimensions, engine data, and SMMT details. The parser handles UKVD's nested object structure (e.g., `Power: {Bhp: 150, Kw: 112}`).

3.4 DVLA Vehicle Enquiry API

Base URL: `https://driver-vehicle-licensing.api.gov.uk/vehicle-enquiry/v1/vehicles`

Auth: `x-api-key` header

Env Variables: FREE_DEVIAL_KEY (working) / PAID_DEVIAL_KEY (broken — 502)

File: src/app/helpers/davlaAPI.ts

Official UK government API. Returns: make, colour, fuelType, engineCapacity, yearOfManufacture, taxStatus, taxDueDate, motStatus, motExpiryDate, co2Emissions, euroStatus, wheelplan, V5C issue date, and export marker.

3.5 DVSA MOT History API

Base URL: <https://history.mot.api.gov.uk/v1/trade/vehicles>

Auth: OAuth2 client credentials flow

Env Variables: MOT_CLIENT_ID, MOT_CLIENT_SECRET, MOT_API_KEY

Official government MOT history API. Returns full MOT test history with test dates, results, mileage readings, advisories, and failure reasons. Uses OAuth2 token-based authentication.

4. Full Vehicle Report — All 22 Sections

The system returns a comprehensive vehicle history report with 22 sections. Below shows each section, which API provides the data, the current status, and what data is returned.

#	Report Section	Data Source	Status	Fields Returned
1	Tax & MOT Status	DVLA + CCD	WORKING	taxStatus, taxDueDate, taxDaysLeft, motStatus, motExpiryDate, motDaysLeft
2	Vehicle Details	DVLA + CCD + RapidAPI	WORKING	make, model, variant, colour, fuel, transmission, driveType, engineCapacity, year, age, registrationDate, V5C, euroStatus, ULEZ, typeApproval, wheelPlan, bodyStyle, doors, seats, gears
3	Mileage Information	CCD + RapidAPI	WORKING	lastMotMileage, mileageIssues, averageMileage, status, estimatedCurrent, mileageHistory[]
4	MOT History Summary	CCD + DVSA MOT	WORKING	totalTests, passed, failed, passRate
5	Performance	UKVD + CCD	EMPTY*	powerKw, powerBhp, maxSpeedMph, maxTorqueNm, maxTorqueRpm, zeroTo60Mph, peakPower, maxSpeedKph
6	Dimensions & Weight	UKVD + CCD	EMPTY*	width, height, length, wheelBase, kerbWeight, grossWeight, unladenWeight, fuelTankCapacity, numberOfAxles
7	Fuel Economy	UKVD + CCD	EMPTY*	urbanMpg, extraUrbanMpg, combinedMpg
8	Road Tax & CO2	DVLA + CCD	WORKING	cost12Months, cost6Months, co2Emissions, co2EmissionBand
9	Vehicle Flags (Summary)	CCD + RapidAPI	WORKING	exported, safetyRecalls, financeRecord, policeStolen, salvageHistory, writeOff, highRisk, v5cCount, totalKeepers, colourChange, plateChange, imported, scrapped
10	Stolen Vehicle Check	CCD + RapidAPI	WORKING	status, message
11	Write-Off / Damage Report	CCD + RapidAPI	WORKING	status, insurerName, causeOfDamage, lossDate, damageLocations[]

12	Finance Check	CCD + RapidAPI	WORKING	status, message, agreementNumber, companyName, agreementType, agreementDate
13	Salvage History	CCD + RapidAPI	PARTIAL	found, category, mileage, retailValue, salvageLocation, saleDate
14	Ex-Taxi / NHS / Police	CCD + RapidAPI	WORKING	status, message
15	Vehicle Valuation	CCD + RapidAPI	PARTIAL	onTheRoad, dealerForecourt, tradeRetail, tradeAverage, tradePoor, partExchange, privateClean, privateAverage, auction
16	Previous Keepers & V5C	CCD + RapidAPI	PARTIAL	totalKeepers, currentKeeperDuration, keepers[] with dates
17	Plate Changes	RapidAPI	PARTIAL	totalChanges, changes[] with dates and types
18	Colour Changes	RapidAPI	WORKING	status, message
19	Engine Data (Detailed)	UKVD + CCD	EMPTY*	cylinders, fuelSystem, aspiration, arrangement, camType, valvesPerCyl, engineLocation
20	SMMT Details	UKVD + CCD	EMPTY*	smmtMarque, smmtRange, modelVariant, countryOfOrigin, driveType, gearbox, numberOfGears
21	Running Costs	UKVD + CCD	EMPTY*	tax6Months, tax12Months, fuelCost12kMiles, fullTankCost, motCost
22	CO2 Emissions	DVLA + CCD	WORKING	co2Gkm, emissionBand

*** EMPTY sections (5, 6, 7, 19, 20, 21) require either a CCD Live API key or a UKVD production key. These sections have full code integration and parsers ready — they will populate automatically once a working API key is configured.**

5. All API Endpoints

Base URL: `http://localhost:5000/api/v1`

5.1 Authentication

Endpoint	Method	Body / Params	Note
<code>/auth/register</code>	POST	{ fullName, email, password }	No auth required
<code>/auth/login</code>	POST	{ email, password }	Returns accessToken
<code>/auth/forgot-password</code>	POST	{ email }	Sends OTP email
<code>/auth/verify</code>	POST	{ email, otp }	Verify OTP
<code>/auth/reset-password</code>	POST	{ email, newPassword }	After OTP verified
<code>/auth/change-password</code>	POST	{ oldPassword, newPassword }	Bearer token required

5.2 Vehicle Check (DVLA)

Endpoint	Method	Body / Params	Note
<code>/check-car/check</code>	POST	{ registrationNumber }	Auto free/paid based on subscription
<code>/check-car/free</code>	POST	{ registrationNumber }	Always uses free DVLA key
<code>/check-car/paid</code>	POST	{ registrationNumber }	Always uses paid DVLA key
<code>/check-car/mot-history</code>	POST	{ registrationNumber }	DVSA + DVLA combined
<code>/check-car/my-checkcar</code>	GET	Query: page, limit	Paginated list of user checks
<code>/check-car/single/:id</code>	GET	Path: MongoDB _id	Single check result
<code>/check-car/single/:id</code>	DELETE	Path: MongoDB _id	Delete a check

5.3 CarTax (Full Report — 22 Sections)

Endpoint	Method	Body / Params	Note
<code>/car-tax/check</code>	POST	{ vrm }	Full vehicle report — FREE for all users
<code>/car-tax/check-pdf</code>	POST	{ vrm }	Full check + PDF download (PAID only)
<code>/car-tax/report-pdf/:id</code>	GET	Path: MongoDB _id	Download PDF for existing report
<code>/car-tax/my-reports</code>	GET	Query: page, limit	Paginated list of user reports
<code>/car-tax/single/:id</code>	GET	Path: MongoDB _id	Single report
<code>/car-tax/single/:id</code>	DELETE	Path: MongoDB _id	Delete report

5.4 Subscription & Payment

Endpoint	Method	Body / Params	Note
/subscribe	GET	Query: page, limit	List all subscription plans
/subscribe	POST	{ planName, price, features }	Admin: create plan
/subscribe/:id	PATCH	{ planName, price, ... }	Admin: update plan
/payment/:subscribeld	POST	Path: plan ID	Create Stripe PaymentIntent
/webhook	POST	Stripe event payload	Stripe webhook handler

5.5 Other Endpoints

Endpoint	Method	Body / Params	Note
/user/profile	GET	—	Get authenticated user profile
/user/profile	PUT	form-data (multipart)	Update profile with avatar
/user	GET	Query: page, limit	Admin: list all users
/dashboard/overview	GET	—	Admin: dashboard stats
/dashboard/chart	GET	Query: year	Admin: revenue chart data
/contact	POST	{ fullName, email, message }	Contact form submission
/mot-history/my	GET	—	Get user's MOT history checks

6. API Keys & Environment Configuration

All API keys are stored in the `.env` file at the project root. Below is the complete list of required environment variables for the vehicle check system.

Env Variable	Status	Service	Notes
<code>FREE_DEVIAL_KEY</code>	WORKING	DVLA Free	Basic DVLA data, limited rate
<code>PAID_DEVIAL_KEY</code>	BROKEN	DVLA Paid	UUID format key — 502 auth error
<code>CARTAX_API_KEY</code>	WORKING	RapidAPI	Comprehensive vehicle reports
<code>UKVD_API_KEY</code>	NEEDS TEST	UKVD Direct	New key: 0DBC4CC1... (just added)
<code>CCD_TEST_API_KEY</code>	WORKING*	CheckCarDetails	Test key, limited data points
<code>CCD_LIVE_API_KEY</code>	EMPTY	CheckCarDetails	Not yet purchased (£20/month)
<code>MOT_CLIENT_ID</code>	WORKING	DVSA MOT	OAuth2 client ID
<code>MOT_CLIENT_SECRET</code>	WORKING	DVSA MOT	OAuth2 client secret
<code>MOT_API_KEY</code>	WORKING	DVSA MOT	API key for MOT history

* `CCD_TEST_API_KEY` works for `vehicleregistration`, `mot`, and `mileage` endpoints only. Premium endpoints (`Vehiclespecs`, `ukvehicledata`, `carhistorycheck`, `vehiclevaluation`) return 403 and require the live key.

Other Required Environment Variables

- `MONGO_URI` — MongoDB Atlas connection string
- `JWT_SECRET` / `ACCESS_TOKEN_SECRET` / `REFRESH_TOKEN_SECRET` — JWT signing keys
- `STRIPE_SECRET_KEY` / `STRIPE_PUBLISHABLE_KEY` / `STRIPE_WEBHOOK_SECRET` — Stripe payment
- `EMAIL_HOST` / `EMAIL_PORT` / `EMAIL_ADDRESS` / `EMAIL_PASS` — SMTP for OTP emails
- `CLOUDINARY_CLOUD_NAME` / `CLOUDINARY_API_KEY` / `CLOUDINARY_API_SECRET` — Image uploads

7. Subscription & Payment Flow

How Subscription Works

1. Admin creates subscription plans via POST /subscribe (e.g., Premium at £9.99/month).
2. User selects a plan. Frontend calls POST /payment/:subscribeId which returns a Stripe clientSecret.
3. Frontend completes Stripe payment using the clientSecret.
4. Stripe fires a webhook to POST /webhook with payment_intent.succeeded event.
5. Webhook handler adds the user's ID to the Subscribe.user[] array and marks payment as 'completed'.
6. All subsequent car checks automatically use the paid tier. The response includes keyType: 'paid' so the frontend knows which tier was used.

Subscription Check Logic

In car-tax.service.ts, isSubscribed() checks if the user's ID exists in any Subscribe document's user[] array. If subscribed, keyType is set to 'paid' which enables PDF downloads.

8. Current Status — What Works & What's Missing

Fully Working

- User registration, login, JWT authentication (access + refresh tokens)
- Free DVLA car check (POST /check-car/free) — works with all UK plates
- Smart check (POST /check-car/check) — auto-selects key based on subscription
- CarTax full report (POST /car-tax/check) — 4-step pipeline with CCD + RapidAPI + UKVD + DVLA
- MOT History (POST /check-car/mot-history) — full DVSA data with test details
- PDF report generation with pdfkit (POST /car-tax/check-pdf for paid users)
- Subscription plans (CRUD for admin)
- Stripe payment integration with webhook
- Report storage in MongoDB with pagination
- Admin dashboard with revenue charts

Currently Empty (Needs API Key)

The following 6 sections appear as empty headers in the report because no API is currently returning this data. The code, parsers, and database schemas are all fully built and ready — they will auto-populate once a working API key is configured:

- Performance (powerBhp, torque, 0-60, top speed)
- Fuel Economy (urban, extra-urban, combined MPG)
- Dimensions & Weight (height, width, length, weight)
- Engine Data (cylinders, aspiration, fuel system)
- SMMT Details (marque, range, model variant, country of origin)
- Running Costs (annual tax, fuel cost, MOT cost)

Known Issues

Issue	Description	Resolution
PAID_DEVIAL_KEY 502	UUID format key causes 502 auth error	Get a valid DVLA paid key from DVLA API portal
CCD Premium 403	Test key has no access to premium data points	Purchase CCD Live key (£20/month)
UKVD New Key	Key 0DBC4CC1... just added, untested	Restart server and test with K5HHV
CCD_LIVE_API_KEY empty	No live key configured	Complete CCD payment signup

9. What's Needed to Get Full Data

To fill all 22 sections with complete data, one of the following actions is needed:

Option A: CCD Live API Key (Recommended)

Purchase a CheckCarDetails Live API key for £20/month. This single key unlocks ALL 7 data points including Vehiclespecs (performance, fuel economy, dimensions), ukvehicledata (SMMT, engine data), carhistorycheck (stolen, finance, write-off, salvage), and vehiclevaluation.

1. Go to <https://api.checkcardetails.co.uk> and sign in
2. Add a payment method (credit/debit card)
3. Subscribe to the Live API plan (£20/month)
4. Copy the Live API key from the dashboard
5. Add to .env: `CCD_LIVE_API_KEY=<your_key>`
6. Request premium data point access from CCD support if not automatically enabled
7. Restart the server — all 22 sections should now populate

Option B: UKVD Production Key

If the newly added UKVD key (0DBC4CC1...) is a production key, it should already provide performance, fuel economy, dimensions, engine data, and SMMT details. Test by restarting the server and checking K5HHV.

If it's still a sandbox key (returns 401), you need to upgrade your UKVD account at <https://www.ukvehicledata.co.uk> and get a production key.

Option C: Both Keys (Best Coverage)

For maximum data coverage and redundancy, configure both CCD Live + UKVD production keys. The 4-step merge pipeline ensures that if one API misses a field, the other fills it in.

10. Key Files & Project Structure

File	Purpose
src/app/helpers/checkCarDetailsAPI.ts	CCD API integration — <code>fetchAllCcdData()</code> calls 7 endpoints in parallel, <code>parseCcdResponse()</code> maps to unified schema
src/app/helpers/carTaxAPI.ts	RapidAPI + UKVD integration — <code>callCarTax()</code> , <code>callUkvd()</code> , <code>parseRapidApiResponse()</code> , <code>parseUkvdData()</code> with nested object handling, <code>mergeReports()</code> , <code>enrichWithDvla()</code>
src/app/helpers/dvlaAPI.ts	DVLA API — <code>freeDVLACarCheck()</code> and <code>paidDVLACarCheck()</code>
src/app/helpers/pdfReport.ts	PDF generation with <code>pdfkit</code> — <code>generateVehicleReportPdf()</code> creates A4 PDF with all 22 sections
src/app/module/car-tax/car-tax.service.ts	Main service — 4-step pipeline: CCD → RapidAPI → UKVD → DVLA with deep merge
src/app/module/car-tax/car-tax.controller.ts	Controller — <code>/check</code> (free JSON), <code>/check-pdf</code> (paid PDF), <code>/report-pdf/:id</code> , <code>/my-reports</code> , <code>/single/:id</code>
src/app/module/car-tax/entities/car-tax.entity.ts	Mongoose schema with 22 nested sub-schemas for the full report
src/app/config/index.ts	Central config — reads all API keys from <code>.env</code> , auto-detects CarTax provider
.env	Environment variables — all API keys, MongoDB URI, JWT secrets, Stripe keys

11. Testing Guide

Quick Test with Postman / cURL

8. Register: POST /api/v1/auth/register with { fullName, email, password }
9. Login: POST /api/v1/auth/login → copy the accessToken from response
10. Set header: Authorization: Bearer <token>
11. Full report: POST /api/v1/car-tax/check with { vrm: 'K5HHV' }
12. DVLA check: POST /api/v1/check-car/check with { registrationNumber: 'K5HHV' }
13. MOT history: POST /api/v1/check-car/mot-history with { registrationNumber: 'K5HHV' }

Test Number Plates

For the CCD test key and UKVD sandbox, use plates containing the letter 'A': AA19AAA, AB12CDE, BA51CAT, VA12ABC. For DVLA free key, any valid UK plate works (e.g., K5HHV).

Verifying Data Flow

The server console logs every step. After calling /car-tax/check, look for:

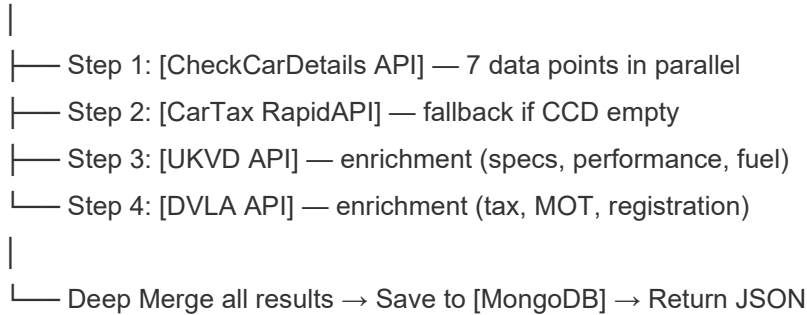
- [CarTax] Step 1: Fetching CheckCarDetails data for K5HHV
- [CarTax] Step 2: Fetching RapidAPI report (fallback) — only if Step 1 had no make
- [CarTax] Step 3: Fetching UKVD data (new key) — should show performance/fuel/dimensions
- [CarTax] Step 4: Fetching DVLA data
- [CarTax] FINAL RESULT — shows what data was collected

12. Architecture Diagram

Request Flow

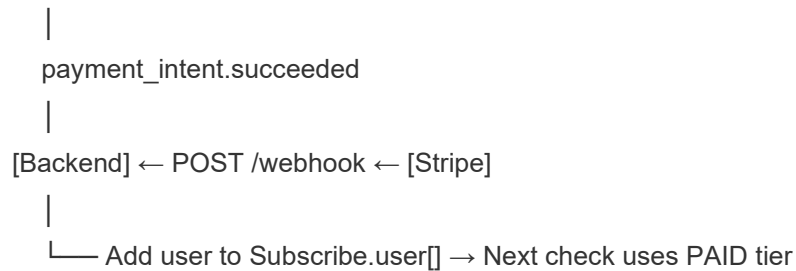
Frontend App → NestJS Backend (JWT Auth) → Multiple APIs → MongoDB → JSON/PDF Response

[Frontend] — POST /car-tax/check → [NestJS Backend]



Subscription Flow

[User] — POST /payment/:planId → [Backend] → [Stripe]



13. Sample JSON Response Structure

Below is the full JSON structure returned by POST /car-tax/check. All 22 sections are always present; fields are null/undefined when the API does not return them.

```
{  "status": {    "taxStatus", "taxDueDate", "taxDaysLeft", "motStatus", "motExpiryDate", "motDaysLeft"  },  "vehicleDetails": {    "make", "model", "modelVariant", "colour", "fuelType", "transmission", "driveType", "engineCapacity", "yearOfManufacture", "vehicleAge", "dateFirstRegistered", "euroStatus", "ulezCompliant", "typeApproval", "wheelPlan", "bodyStyle", "numberOfDoors", "numberOfSeats", "numberOfGears"  },  "mileage": {    "lastMotMileage", "mileageIssues", "averageMileage", "mileageStatus", "estimatedCurrentMileage", "mileageHistory[]"  },  "motHistory": {    "totalTests", "passed", "failed", "passRate"  },  "performance": {    "powerKw", "powerBhp", "maxSpeedMph", "maxTorqueNm", "zeroTo60Mph"  },  "dimensions": {    "widthMm", "heightMm", "lengthMm", "kerbWeightKg", "grossWeightKg"  },  "fuelEconomy": {    "urbanMpg", "extraUrbanMpg", "combinedMpg"  },  "roadTax": {    "cost12Months", "cost6Months", "co2Emissions", "co2EmissionBand"  },  "vehicleFlags": {    "exported", "policeStolen", "financeRecord", "writeOff", "salvageHistory", "safetyRecalls"  },  "stolenCheck": {    "status", "message"  },  "writeOffReport": {    "status", "insurerName", "causeOfDamage"  },  "financeReport": {    "status", "message", "companyName", "agreementType"  },  "salvageHistory": {    "found", "category"  },  "exTaxiCheck": {    "status", "message"  },  "valuation": {    "dealerForecourt", "tradeRetail", "privateClean", "auction"  },  "previousKeepers": {    "totalKeepers", "keepers[]"  },  "plateChanges": {    "totalChanges", "changes[]"  },  "colourChanges": {    "status", "message"  },  "engineData": {    "cylinders", "fuelSystem", "aspiration", "arrangement"  },  "smmtDetails": {    "smmtMarque", "smmtRange", "modelVariant", "countryOfOrigin"  },  "runningCosts": {    "tax12Months", "fuelCost12kMiles", "fullTankCost"  },  "emissions": {    "co2Gkm", "emissionBand"  } }
```

14. Summary & Next Steps

What's Done

- Complete 4-step API pipeline with intelligent fallback and deep merge
- All 22 report sections coded with parsers for CCD, RapidAPI, UKVD, and DVLA
- PDF report generation for paid/subscribed users
- Free users see full JSON data; paid users can also download PDF
- Stripe subscription with webhook integration
- MongoDB storage with full schemas for all 22 sections
- Detailed console logging for debugging data flow

Immediate Next Steps

7. Test UKVD new key: Restart server, call POST /car-tax/check with { vrm: 'K5HHV' }, check console for Step 3 output.
8. If UKVD key works: Performance, Fuel Economy, Dimensions, Engine Data, and SMMT sections will now have data.
9. If UKVD key fails (401): Upgrade UKVD account to production or purchase CCD Live key.
10. Purchase CCD Live API key (£20/month) for maximum data coverage across all 22 sections.
11. Fix DVLA paid key: Get a valid DVLA paid API key (current one is UUID format which causes 502).